

Experiment :- 4

AIM :- Write About Software Development Standards used in Industries, including Both: Development Standards & Documentation Standards used in the minor project "BizFlow - Inventory and Billing Management System".

Software Development Standards are a set of rules, guidelines, and best practices followed during software development.

These standards help developers create software that is:

- Reliable
- Secure
- Efficient
- Easy to maintain

Documentation Standards help in maintaining proper project records, requirements, design details, testing reports, and user manuals.

Industries use these standards to improve software quality, maintainability, and teamwork.

1. Software Development Standards

Software Development Standards are guidelines followed during coding, designing, testing, and maintaining software.

Types of Software Development Standards

i) Coding Standards

Coding standards define how source code should be written.

Rules Included:

- Proper variable and function naming
- Correct indentation
- Meaningful comments
- Code formatting
- Consistent folder structure

Advantages:

- Easy to read code
- Easy debugging

- Better teamwork
- Improved maintainability

ii) Design Standards

Design standards define how software architecture and system design should be created.

Includes:

- UML Diagrams
- Database Design
- System Architecture
- Modular Design
- User Interface Design

Advantages:

- Better understanding of system
- Easier maintenance
- Improved scalability
- Better project planning

iii) Testing Standards

Testing standards ensure software works correctly and meets requirements.

Types of Testing:

1. Unit Testing
2. Integration Testing
3. System Testing
4. User Acceptance Testing

Advantages:

- Reduces errors
- Improves quality
- Increases reliability
- Ensures system stability

iv) Security Standards

Security standards protect software and business data from unauthorized access.

Includes:

- User Authentication
- Secure Login Mechanism
- Password Protection
- Role-Based Access Control
- Database Security

Advantages:

- Prevents unauthorized access
- Protects business data
- Improves system security
- Enhances user trust

v) Quality Standards

These standards ensure software quality and reliability.

Common Standards:

- ISO 9001
- IEEE Standards
- Software Quality Assurance Practices
- Code Review Standards

Advantages:

- Improves software quality
- Increases customer satisfaction
- Enhances reliability
- Reduces software defects

2. Documentation Standards

Documentation Standards define how project documents should be prepared and maintained.

Types of Documentation Standards

i) SRS (Software Requirement Specification)

SRS contains all software requirements.

Includes:

- Functional Requirements
- Non-Functional Requirements
- Hardware Requirements
- Software Requirements

Advantages:

- Clear project understanding
- Reduces confusion
- Improves communication among stakeholders

ii) Design Documentation

Contains system design details.

Includes:

- UML Diagrams
- Database Schema
- System Architecture
- Module Design

Advantages:

- Easy implementation
- Better project planning
- Easier future modifications

iii) Testing Documentation

Contains testing details and results.

Includes:

- Test Cases
- Bug Reports
- Testing Reports
- Validation Results

Advantages:

- Tracks software quality
- Helps debugging
- Improves software reliability

iv) User Documentation

Helps users understand software usage.

Includes:

- User Manual
- Installation Guide
- Help Documentation
- Feature Descriptions

Advantages:

- Easy software usage
- Better user experience
- Reduced training requirements

v) Maintenance Documentation

Contains future update and maintenance information.

Includes:

- Bug Fix Reports
- Version Updates

- Change Logs
- Feature Enhancement Records

Advantages:

- Easy future maintenance
- Tracks software changes
- Improves long-term support

Importance of Software Standards

i) Improves Software Quality

Standards reduce software errors and improve reliability.

ii) Better Team Coordination

Developers follow common rules and practices.

iii) Easy Maintenance

Well-structured software is easy to update and enhance.

iv) Increased Security

Security standards protect business and customer data.

v) Better Documentation

Project information remains organized and accessible.

Standards Used in BizFlow Inventory and Billing Management System

In the BizFlow Inventory and Billing Management System project, the following standards were used:

Development Standards

- JavaScript Coding Standards
- React Development Standards
- Modular Programming
- Database Design Standards
- Testing Standards
- Security Practices

Documentation Standards

- SRS Documentation
- Use Case Documentation
- System Design Documentation
- Testing Reports
- User Documentation